Mavlyudabegim Mekhrilloeva

Working Student – AI, Automation & Data | M.Sc. Computer Science, Saarland University

Saarbrücken, Germany · +49 157 50730857 · mekhrullaeva@gmail.com

LinkedIn · GitHub

Available immediately · up to 20 hours/week · open to relocation

Profile

M.Sc. Computer Science student (Machine Learning focus) at Saarland University and DAAD scholar with 2 years of professional experience as a software engineer building reporting tools for insurance companies and financial institutions. Skilled at translating business requirements into practical technical solutions and explaining technical concepts to non-technical stakeholders; passionate about applying AI and automation to real business problems.

Skills

AI & Data	Machine Learning, Neural Networks, NLP / Machine Translation, Reinforcement Learning, data analysis, data quality, PyTorch, NumPy, pandas
Programming & Data	Python, SQL, PostgreSQL, TypeScript/JavaScript
Automation & Tools	Django (ORM), Angular (NgRx), Git, Docker, Kubernetes, AWS, SonarQube, Jira
Quality & Testing	Unit and end-to-end testing (Jest), automated code analysis, debugging
Office	MS Office (Excel, Word, PowerPoint, Outlook)
Soft Skills	Stakeholder communication, requirements analysis, analytical problem-solving, teamwork, independent work

Professional Experience

Software Engineer (Python / Angular) — Exadel 06/2022 – 04/2024 · Tashkent, Uzbekistan

- Developed a web-based reporting platform for insurance companies and financial institutions to build regulatory and financial reports from ready-made building blocks.
- Translated client business requirements into technical solutions by tailoring core form designs into custom financial reports with spreadsheet-like formulas and calculation features.
- Automated report generation with Python/Django services on a PostgreSQL database, producing Excel and PDF documents, reducing manual work for business users.
- Built data-heavy Angular features with NgRx state management (virtual scroller for large datasets, financial report tables), improving usability by 65%.
- Improved data quality and reliability: used SonarQube for automated code analysis and fixed validation, authorization and API errors on server and UI side; wrote unit and end-to-end tests with Jest decreasing bugs in production by 70% .
- Collaborated with stakeholders in an international, cross-functional team of 15 people.

Education

M.Sc. Computer Science (Informatik) — Saarland University, Saarbrücken 10/2024 – expected 01/2027

- All coursework completed; thesis in progress: *Delegated Classification — contract design for ML as a Stackelberg game*
- Relevant courses: Machine Learning, Neural Networks, Machine Translation (NLP), Optimization Algorithms in ML, Cybersecurity in ML
- DAAD Scholarship holder (All Disciplines)

B.Sc. Computer Engineering — Tashkent University of Information Technologies 2018 – 2022

- Grade: 1.1 (German grading scale) · Thesis: *Algorithms for preprocessing EMG biosignals*
- Relevant courses: Data Structures and Algorithms, Machine Learning, Artificial Intelligence, Numerical Methods

Projects

- **Scribble-Based Image Segmentation** (Python, PyTorch) — built and evaluated a machine learning prototype for foreground/background segmentation (80/20 split, early stopping), reaching 95% mean IoU on held-out data.
- **Q-Learning Tic-Tac-Toe Agent** (Python, Reinforcement Learning) — implemented a self-learning agent, achieving a 92% win/draw rate.
- **Delegated Classification** (Master's thesis) — modelling incentive contracts between a principal and an ML agent as a Stackelberg game; developing robust contracts under delegation.

Languages & Awards

Languages: English (C1, fluent) · German (B1, attending language course) · Russian (C1) · Uzbek (native)

Awards: DAAD Scholarship — All Disciplines (2024)